

Question Paper

Exam Date & Time: 30-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025

Embedded programming [ECE_4450]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Suggest the design requirements in Embedded Systems and explain each of those. (4)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
B) Discuss the importance of Compiling and Linking in Embedded Systems. (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) Elaborate the role of ROM Emulators in Embedded Systems. (3)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Discuss the classification system of Memory in Embedded Systems with a brief note on Flash memory characteristics. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Elaborate the selection criteria used in case of RTOS. (3)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
C) Discuss the different techniques used to measure the code efficiency. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) With a neat schematic, explain the architecture of a Bus. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Elaborate on the importance of Interrupts used in an Embedded OS with relevant diagram. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Extend the significance of Built-in functions used in a Microprocessor. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Discuss the salient features of Round robin architecture without interrupts and its drawbacks. (4)
A) Course Outcome: "CO4" / Cognitive Level: "Application"

B) With a neat sketch, discuss the different task states used in RTOS. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

C) Elaborate the execution flow in Semaphores by mentioning suitable example. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

5) With a neat diagram, explain the process building software used in Embedded Systems. (4)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) What would be the poor plan for testing? Explain using a pseudocode. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

C) Discuss the significance of Instruction set simulators. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECE_4450:Embedded programming

#	Course Outcome	Description	Marks
1	1	CO 1	4
2	2	CO 2	6
3	3	CO 3	20
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECE_4450:Embedded programming

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	10(20%)
3	Application	8	27(54%)
4	Analysis	4	13(26%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)